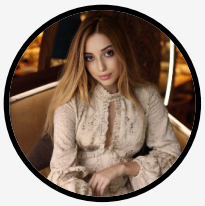
europass



Anastasiia Stepanenko

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LinkedIn: anastasiia-stepanenko-1a0871327

Location: Munich, Germany | **GitHub:** nastya-star

ABOUT ME

M.Sc. Mathematics in Data Science student at the Technical University of Munich, with a background in systems analysis and numerical methods. Developed a diet-optimization web application comparing linear, robust and stochastic programming approaches. Research experience includes penalty methods, two-sided constraints and AMPL/HiGHS implementation. Presented optimization research at Ukrainian Mathematics Days 2026, Institut Henri Poincaré, Paris.

EDUCATION & TRAINING

01/09/2021 - 30/08/2025 - KYIV, UKRAINE

B.SC. IN SYSTEMS ANALYSIS AND CONTROL- NATIONAL TECHNICAL UNIVERSITY OF UKRAINE "IGOR SIKORSKY KYIV POLYTECHNIC INSTITUTE"

Relevant coursework: operations research; linear and nonlinear optimization; probability and statistics; numerical methods; algorithms and data structures; databases; time-series analysis.

Bachelor's thesis:

Design and Implementation of an Information System for the Cost-Optimized Formulation of Feline Diets in Compliance with Nutritional Standards

- Implemented and compared linear programming, robust optimization and stochastic programming approaches for minimum-cost diet formulation under nutritional constraints.
- Processed food-composition data and modelled uncertainty in nutrient coefficients; used UMAP and t-SNE to explore similarities between foods.
- Integrated optimization models and visualizations into a web application for diet planning.

Field of study: Systems analysis | **Final grade:** 92

Qualification: EQF Level 6 | **Credits:** 240 ECTS

Website: https://kpi.ua/

Link https://ela.kpi.ua/items/5b2b7c9f-a9eb-41b4-8cd4-d797b283ea9e

01/03/2022 - 31/08/2022 - MITTWEIDA, GERMANY

NON-DEGREE INTERNATIONAL VISITING STUDENT- HOCHSCHULE MITTWEIDA - UNIVERSITY OF APPLIED SCIENCES

- Completed university-level coursework in calculus, linear algebra, probability, statistics, numerical methods, mathematical modelling, optimization, and scientific programming with Python.

Field(s) of study: Statistics , Mathematics | **Level in EQF:** 6 | **Website:** https://www.hs-mittweida.de/

01/03/2022 - 31/08/2022 - CHEMNITZ, GERMANY

DSH-2 (DEUTSCHE SPRACHPRÜFUNG FÜR DEN HOCHSCHULZUGANG)- STUDIENKOLLEG

HOCHSCHULE MITTWEIDA

- Achieved DSH-2, demonstrating German-language proficiency for university study at C1 level.

Field(s) of study: Academic German | **Website:** https://studienkolleg-hs-mittweida.de/

01/09/2025 - Current - KYIV, UKRAINE

M.SC. MATHEMATICS- KYIV ACADEMIC UNIVERSITY

Academic focus: mathematical optimization, numerical methods, probability, stochastic processes, statistical learning and computational mathematics.

Computational methods: scientific programming with Python; numerical optimization; stochastic modelling; regularization and model selection.

Research experience: linear programs with two-sided constraints, penalty formulations, the r-algorithm and AMPL/HiGHS implementation.

Research interests: robust, stochastic and nonsmooth optimization; large-scale mathematical programming; optimal control.

Field of study: Mathematics | **Qualification:** EQF Level 7

Type of credits: ECTS | **Website:** https://kau.org.ua/

10/2026 - Present | Expected completion: 09/2028 | MUNICH, GERMANY

M.SC. MATHEMATICS IN DATA SCIENCE- TECHNICAL UNIVERSITY OF MUNICH (TUM)

Current studies: Statistical Learning, Compressed Sensing, Foundations in Data Engineering, Artificial Intelligence in Medicine and Distributed Database Web Applications.
Focus: mathematical optimization, data science and decision-making under uncertainty.

Level in EQF: 7

07/09/2026 - 11/09/2026 - SAN MINIATO, ITALY

SUMMER SCHOOL — NONLINEAR AND STRUCTURED OPTIMIZATION: THEORY AND ALGORITHMS- MATHEMATICS RESEARCH CENTER “ENNIO DE GIORGI”, SCUOLA NORMALE SUPERIORE

Academic focus: nonlinear and structured optimization; convex analysis and generalized convexity; smooth and nonsmooth optimization.

Optimality and duality: KKT conditions; subgradients and subdifferentials; Lagrangian and Fenchel duality.

Algorithmic methods: first-order and proximal algorithms; forward–backward and Douglas–Rachford splitting; alternating direction method of multipliers (ADMM).

Selected topics: multiobjective and bilevel optimization; Pareto efficiency and scalarization.

Participation: selected for an international summer school with 16 available places; board and accommodation funded by the organizers.

Website: https://crmdegiorgi.sns.it/

Link https://indico.sns.it/event/148/attachments/370/1045/Syllabus.pdf

PUBLICATIONS

Can Artificial Intelligence Replace the Teacher and Is There a Threat to Students? 2024
Conference abstract examining the capabilities and limitations of AI in education, risks for students and the responsible use of AI-based teaching and learning technologies.

Stepanenko, A. (2024). Can Artificial Intelligence Replace the Teacher and Is There a Threat to Students? In Proceedings of the VI All-Ukrainian Scientific and Practical Conference “Modern Information Technologies in Education and Science”, pp. 223-224.

Link https://www.tsatu.edu.ua/kn/wp-content/uploads/sites/16/uman_15_11_24.pdf

Mathematical Model for Planning a Minimum-Cost Diet 2025

Conference abstract formulating minimum-cost diet planning as a linear program under nutritional constraints, with scope for multi-meal and uncertainty-aware extensions.

Stepanenko, A. (2025). Mathematical Model for Planning a Minimum-Cost Diet. In Proceedings of the XIV International Scientific and Practical Conference “Glushkov Readings”, Kyiv, Ukraine, 30 May 2025, pp. 331-333.

The First Diet Problem 2026

First author of a Ukrainian-language popular-science article explaining Stigler’s diet problem, its linear programming formulation and the historical application of the simplex method to minimum-cost nutrition planning.

Stepanenko, A., Korablov, M., & Stetsyuk, P. (2026). Перша задача про дієту The First Diet Problem . Країна знань Kraina Znan , No. 9. Popular-science article; Ukrainian.
Read the published article

CONFERENCES & SEMINARS

28/11/2025 - 29/11/2025 Online

A Penalty & r-Algorithm Approach for Linear Programs with Many Two-Sided Constraints

Presented a penalty-based approach to linear programs with many two-sided constraints. Investigated the r-algorithm and its application to large constraint sets, focusing on computational implementation and comparison with established optimization solvers.

Implemented the method in GNU Octave and compared runs with GLPK and selected NEOS solvers. Examined formulation choices and numerical behaviour on the tested instances.

Project materials: LinkedIn project portfolio

Methods: linear programming; two-sided constraints; penalty formulations; numerical optimization.

Tools: GNU Octave, GLPK and NEOS.

13/02/2026 Online

Diet Problems: A Review of Classical and Modern Solution Methods

Presented a seminar tracing diet optimization from Stigler’s linear programming formulation to mixed-integer, robust, stochastic and multi-objective models. Reviewed two-sided nutrient constraints, acceptability, sustainability and data-driven approaches to menu planning.

Format: research seminar | **Location:** online

04/06/2026 - 05/06/2026 Institut Henri Poincaré, Paris, France

Ukrainian Mathematics Days 2026

Presented **“An Adaptive Nutrient Requirements Optimization Model for the Multi-Meal Diet”** as a poster and a three-minute talk at the Institut Henri Poincaré. Extended minimum-cost diet planning with meal-indexed variables and nutrient constraints across multiple meals, incorporating meal timing and training-related requirements. Communicated the model and its motivation to an international mathematical audience.

Link https://ukrmathday2026.sciencesconf.org/resource/page/id/4

SKILLS

Mathematical optimization

Linear Programming | Constrained Optimization | Mathematical Modelling | Operations Research | Penalty Methods | Numerical Optimization | Robust Optimization | Stochastic Programming

Mathematical foundations

Mathematical Analysis | Linear Algebra | Stochastic Processes | Probability Theory | Differential Equations | Mathematical Statistics | Numerical Linear Algebra | Time-Series Analysis

Programming and optimization tools

Python | NumPy | pandas | SciPy | SQL | AMPL | HiGHS | GNU Octave / MATLAB | C++ | Matplotlib | Jupyter | Git | Linux | LaTeX

CERTIFICATIONS

11/05/2025 University of Alberta
Reinforcement Learning Specialization

Online - Studied prediction and control in sequential decision problems, including temporal-difference learning, Monte Carlo methods, function approximation, policy evaluation, and reinforcement-learning systems under uncertainty.

Link https://coursera.org/share/4cd4307846f3ba0391a6ac9a2007da9d

31/01/2025 The Johns Hopkins University
Advanced Statistics for Data Science

Online - Studied least squares, linear models, statistical inference, and regression analysis; completed a housing-price project involving data cleaning, exploratory analysis, predictor comparison, and model interpretation.

Link https://www.coursera.org/account/accomplishments/specialization/MYE01J0YZS87

12/09/2024 SAS
Analyzing Time Series and Sequential Data Specialization

Online - Studied feature engineering, modelling, diagnostics, and forecasting for sequential and time-dependent data, with practical applications to housing and price datasets.

Link https://coursera.org/share/16c1fe063d565511d586d3c705200306

14/09/2024 IBM
IBM Data Science Professional Certificate

Online - Completed project-based training in Python, SQL, data preparation, visualization, statistical analysis, and machine learning, including workflows for transforming raw data into reproducible analytical results.

Link https://coursera.org/share/4e05005110b041b99e088db2b44aa682

09/09/2024 University of Michigan
Foundational Finance for Strategic Decision Making Specialization

Online - Studied financial decision-making, valuation, stocks and bonds, the time value of money, risk, and the evaluation of investment alternatives.

Link https://coursera.org/share/ca7e0495013dd2deadc7aab4c9322c9d

24/10/2024 IBM
Applied Data Science Specialization

- Applied Python-based data analysis and visualization methods in practical projects, including data preparation, exploratory analysis, and communication of quantitative findings.

Link https://www.coursera.org/account/accomplishments/specialization/04BHGWI0HYB4

14/09/2024 Microsoft

Microsoft Cybersecurity Analyst Specialization

- Completed training in network security, identity and access management, threat mitigation, security compliance, and cybersecurity solutions within Microsoft Azure.

Link https://www.coursera.org/account/accomplishments/specialization/AWX84TDGV0Y0

LANGUAGE SKILLS

Mother tongue(s): **UKRAINIAN**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C2	C2	C1
GERMAN	B2	C1	C1	C1	C1

VOLUNTEERING

01/05/2022 - CURRENT Chemnitz

• Supported communication in Ukrainian, German and English in medical, social and administrative settings. Interpreted sensitive discussions involving doctors, social workers, volunteers and public institutions while maintaining confidentiality, accuracy and neutrality.

WORK E PERIENCE

22/06/2023 - Present | Germany

INTERPRETER - SprInt AGIUA e.V.

• Provided Ukrainian/Russian-German interpreting in medical, social and administrative settings. Supported accurate communication between clients, healthcare professionals and public institutions, maintaining confidentiality and impartiality.